

INDIAN STATISTICAL INSTITUTE

Linear Statistical Models

(B.Stat. 3rd year)

Semester 1, 2026-27

Homework 2

Due: September 2, 2026

1. Consider the linear model:

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}, \quad \boldsymbol{\varepsilon} \sim N(0, \sigma^2 I_n) \quad (1)$$

where $\mathbf{X}$ is $n \times p$ with $\text{rank}(\mathbf{X}) \leq p$. Define $\boldsymbol{\mu} = \mathbb{E}(\mathbf{Y})$ and $\boldsymbol{\theta} = \mathbf{X}^T \boldsymbol{\mu}$.

- (a) Show that, the least squares estimator of $\boldsymbol{\mu}$ under model (1) is $\hat{\boldsymbol{\mu}} = \mathbf{P}_{\mathbf{X}} \mathbf{Y}$ where $\mathbf{P}_{\mathbf{X}}$ denotes the orthogonal projection onto the column space of $\mathbf{X}$.
- (b) Show that $\boldsymbol{\theta}$ is estimable and its BLUE is given by $\hat{\boldsymbol{\theta}} = \mathbf{X}^T \mathbf{Y}$.
- (c) Show that if $\mathbf{a}^T \boldsymbol{\beta}$ is estimable, then the BLUE of $\mathbf{a}^T \boldsymbol{\beta}$ is $\mathbf{w}^T \hat{\boldsymbol{\theta}}$ for some $\mathbf{w} \in \mathbb{R}^p$.

2. Consider the linear regression model

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}, \quad (2)$$

with $\boldsymbol{\beta} \in \mathbb{R}^p$, $\text{rank}(\mathbf{X}) = p$ and $\boldsymbol{\varepsilon} \sim N_n(0, \sigma^2 I)$. Let $\hat{\boldsymbol{\beta}} = (\hat{\beta}_1, \dots, \hat{\beta}_p)^T$ be the least squares estimator of $\boldsymbol{\beta}$. Let $\mathbf{X}_{\cdot j}$ denote the j -column of $\mathbf{X}$ and let $\mathbf{X}_{-j}$ denote the $n \times (p-1)$ submatrix of $\mathbf{X}$ with the j -th column removed, for $j = 1, \dots, p$.

- (a) Show that $\text{Var}(\hat{\beta}_j) = \sigma^2 / \mathbf{X}_{\cdot j}^T (\mathbf{I} - \mathbf{P}_{-j}) \mathbf{X}_{\cdot j}$ where $\mathbf{P}_{-j}$ denotes the orthogonal projector onto the column space of $\mathbf{X}_{-j}$.
- (b) Now consider the one variable linear regression model (a submodel of the general model described above):

$$\mathbf{Y} = \mathbf{X}_{\cdot j} \beta_j + \boldsymbol{\varepsilon}. \quad (3)$$

If this submodel (3) happens to be true (so that same assumption on $\boldsymbol{\varepsilon}$ holds), then what would be the distribution of $\tilde{\beta}_j$, the least squares estimator of β_j ? If however, the submodel is not true but the full model (2) is true, then what is the distribution of $\tilde{\beta}_j$?

- (c) Give an argument to say why it is preferable to use the submodel (3) for estimating β_j , rather than the full model, if indeed the submodel is correct. However, if the submodel (3) is not correct, state at least one disadvantage of using this model for statistical inference.

- (d) Can you think of any circumstances when the distributions of $\hat{\beta}_j$ (from the full model (2)) and $\tilde{\beta}_j$ (from the submodel (3)) coincide?

3. Consider the linear regression model

$$Y_i = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_{p-1} x_{i(p-1)} + \varepsilon_i, \quad \varepsilon_i \stackrel{i.i.d.}{\sim} N(0, \sigma^2).$$

- (a) Show that the R^2 can be expressed as

$$R^2 = \frac{\tilde{\beta}^T \tilde{\mathbf{X}}^T \tilde{\mathbf{X}} \tilde{\beta}}{\sum_{i=1}^n (Y_i - \bar{Y})^2}$$

where $\tilde{\mathbf{X}}$ is the $n \times (p-1)$ matrix with (i, j) -th element equal to $x_{ij} - \bar{x}_{.j}$ where $\bar{x}_{.j} = \sum_{i=1}^n x_{ij}/n$, and $\tilde{\beta}$ is the BLUE for the vector $(\beta_1, \dots, \beta_{p-1})^T$.

- (b) Find the distribution of $R^2/(1-R^2)$, assuming that the linear regression model above is true.
4. Let I_n denote the identity matrix in $\mathbb{R}^n$. Let $\mathbf{A}$ be an $n \times n$ positive semidefinite matrix such that $\text{rank}(\mathbf{A}) = r$ and $\text{rank}(I_n - \mathbf{A}) = n - r$ for some $1 \leq r < n$. Then prove that $\mathbf{A}$ is an orthogonal projection matrix. In addition, if $Y \sim N(\mu, I_n)$, then show (without using Fisher-Cochran theorem) that if $U_1 := Y^T \mathbf{A} Y$ and $U_2 := Y^T Y - Y^T \mathbf{A} Y$, then U_1 and U_2 are independent, with $U_1 \sim \chi_r^2(\delta_1/2)$ and $U_2 \sim \chi_{n-r}^2(\delta_2/2)$, where $\delta_1 = \mu^T \mathbf{A} \mu$ and $\delta_2 = \mu^T \mu - \mu^T \mathbf{A} \mu$.
5. Consider the model $Y_i = \beta_1 x_{i1} + \beta_2 x_{i2} + \varepsilon_i$, $i = 1, \dots, 8$, where ε_i are uncorrelated with zero mean and variance σ^2 . Use the data given below to estimate $\beta_1, \beta_2, \sigma^2$.

	1	2	3	4	5	6	7	8
Y_i	82	79	74	83	80	81	84	81
x_{i1}	10	9	9	11	11	10	10	12
x_{i2}	15	14	13	15	14	14	16	13

6. Consider the model

$$Y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \beta_{11} x_{i1}^2 + \beta_{22} x_{i2}^2 + \beta_{12} x_{i1} x_{i2} + \varepsilon_i, \quad i = 1, \dots, 7.$$

The predictor variables take the following values.

	1	2	3	4	5	6	7
x_{i1}	1	1	-1	-1	0	0	0
x_{i2}	1	-1	1	-1	0	0	0

Show that $\beta_0, \beta_1, \beta_2, \beta_{11} + \beta_{22}, \beta_{12}$ are estimable and find (nonmatrix) algebraic forms for the estimates of these parametric functions. Find the MSE and the standard errors of the estimates.